

1. Customer Table:

Column Name	Data Type	Description
CustomerID	INT	Primary key
FirstName	VARCHAR(50)	First name of the customer
LastName	VARCHAR(50)	Last name of the customer
Email	VARCHAR(100)	Email address of the customer
Phone	VARCHAR(20)	Phone number of the customer

2. Product Table:

Column Name	Data Type	Description
ProductID	INT	Primary key
Name	VARCHAR(100)	Name of the product
Price	DECIMAL(10,2)	Price of the product
Description	TEXT	Description of produc

3. Order Table:

Column Name	Data Type	Description
OrderID	INT	Primary key
CustomerID	INT	Foreign key referencing Customer table
OrderDate	DATETIME	Date and time of the order
Qty	INT	Quantity of the order
Rate	DECIMAL(10,2)	Product Rate per Qty
TotalAmount	DECIMAL(10,2)	Total amount of the order
ProductID	INT	Foreign key referencing Product table

4. Payment Table:

Column Name	Data Type	Description
PaymentID	INT	Primary key
OrderID	INT	Foreign key referencing Order table
Amount	DECIMAL(10,2)	Amount paid
PaymentDate	DATETIME	Date and time of the payment

1. Create a Store-Procedure insert operation for all tables (Customer, Product, Order, Payment)
2. Create a Store-Procedure Update operation for all tables (Customer, Product, Order, Payment)
3. Create a Store-Procedure Get operation for all tables (Customer, Product, Order, Payment)
4. Create a Store-Procedure Delete operation by table Primary key for all tables (Customer, Product, Order, Payment)
5. Create a stored procedure that updates the price of a product given its ProductID.
6. Write a stored procedure that takes parameters for CustomerID, OrderDate, ProductID, Qty, and Rate, and inserts a new order into the Order table, calculating the TotalAmount based on the Qty and Rate.
7. Design a stored procedure that records a payment for an order by taking OrderID and Amount as parameters and inserting a new record into the Payment table with the provided information.
8. Create a stored procedure that retrieves the total payments made by each customer by joining the Customer and Payment tables and aggregating the amounts for each customer.
9. Write a stored procedure that identifies customers who have not made any payments by comparing the Customer table with the Payment table and returning the relevant records.

- 10.**Develop a stored procedure that calculates the total revenue for a given period by summing up the TotalAmount from the Order table for orders placed within that period.
- 11.**Design a stored procedure that retrieves all orders along with customer and product details by joining the Order, Customer, and Product tables.
- 12.**Retrieve the top N customers with the highest total payments.
- 13.**Retrieve all orders made by customers who have made payments within the last N months.
- 14.**Calculate the total revenue for each product category.
- 15.**Retrieve the most profitable product (highest total revenue).
- 16.**Retrieve customers who have made purchases of a specific product within a given date range.
- 17.**Calculate the average order value for each customer.
- 18.**Retrieve orders with the highest total amounts for each customer.
- 19.**Calculate the total number of orders and the total revenue generated by each customer for a specific year.
- 20.**Retrieve orders that have not been paid within a certain period.
- 21.**Identify customers who have made consecutive purchases within a given timeframe.
- 22.**Calculate the total revenue for each customer in the last N months.
- 23.**Retrieve orders where the product price is higher than the average price of all products
- 24.**Calculate the average time between consecutive orders for each customer.
- 25.**Create a store procedure create with pagination, sorting and searching with order table.